

Deriving Perturbed Ricci Tensor Components

(1)

Recall (for ease):

$$\left\{ \begin{array}{l} \Gamma_{00}^0 = \mathcal{H} + \Psi' \\ \Gamma_{i0}^0 = \partial_i \Psi \\ \Gamma_{00}^i = \partial^i \Psi \\ \Gamma_{jk}^i = -2\delta^i_{(j} \partial_{k)} \Phi + \delta_{jk} \partial^i \Phi \end{array} \right. \quad \Gamma_{ij}^0 = \mathcal{H} \delta_{ij} - [\Phi' + 2\mathcal{H}(\Phi + \Psi)] \delta_{ij}$$

and $R_{\mu\nu} = \partial_\lambda \Gamma_{\mu\nu}^\lambda - \partial_\nu \Gamma_{\mu\lambda}^\lambda + \Gamma_{\lambda\rho}^\lambda \Gamma_{\mu\nu}^\rho - \Gamma_{\mu\lambda}^\rho \Gamma_{\nu\rho}^\lambda$

R_{00} : $R_{00} = \partial_\lambda \Gamma_{00}^\lambda - \partial_0 \Gamma_{0\lambda}^\lambda + \Gamma_{\lambda\rho}^\lambda \Gamma_{00}^\rho - \Gamma_{0\lambda}^\rho \Gamma_{0\rho}^\lambda$
 All contributions of $\lambda=0$ terms in sum cancels

$$\Rightarrow = \partial_i \Gamma_{00}^i - \partial_0 \Gamma_{0i}^i + \underbrace{\Gamma_{i\rho}^i \Gamma_{00}^\rho - \Gamma_{0i}^\rho \Gamma_{0\rho}^i}_{\substack{\Gamma_{i0}^i \Gamma_{00}^0 - \Gamma_{0i}^0 \Gamma_{00}^i + \Gamma_{ij}^i \Gamma_{00}^j - \Gamma_{0i}^j \Gamma_{0j}^i}} \\ \substack{O(2) \quad O(2)}$$

$$= \nabla^2 \Psi - \partial_0 (3(\mathcal{H} - \Phi')) + 3(\mathcal{H} - \Phi')(\mathcal{H} + \Psi') - (\mathcal{H} - \Phi')^2 \delta^i_j \delta^j_i$$

$$\substack{\text{e.g.}} \quad = 3(\mathcal{H} - \Phi')^2 \quad \substack{O(2)} \\ = 3(\mathcal{H}^2 - 2\mathcal{H}\Phi' + \Phi'^2)$$

$$= -3\mathcal{H}' + \nabla^2 \Psi + 3\mathcal{H}(\Phi' + \Psi') + 3\Phi''$$

Roi: $R_{oi} = \partial_\lambda \Gamma_{oi}^\lambda - \partial_i \Gamma_{o\lambda}^\lambda + \Gamma_{\lambda\rho}^\lambda \Gamma_{oi}^\rho - \Gamma_{o\lambda}^\rho \Gamma_{ip}^\lambda$

Terms from $\lambda = 0$:

$$\partial_0 \Gamma_{oi}^0 - \partial_i \Gamma_{oo}^0 + \underbrace{\Gamma_{0\rho}^0 \Gamma_{oi}^\rho}_{\text{cancel}} - \Gamma_{oo}^\rho \Gamma_{ip}^0 + \left[\Gamma_{oj}^0 \Gamma_{oi}^j - \Gamma_{oo}^j \Gamma_{ij}^0 \right]$$

$$= \underbrace{\partial_0 (\partial_i \Psi) - \partial_i (\mathcal{H} + \Psi')}_{\downarrow} + \underbrace{(\partial_i \Psi) [(\mathcal{H} - \Phi') \delta_i^j]}_{(*)} - \underbrace{(\partial^j \Psi) [\mathcal{H} \delta_{ij} - \{\Phi' + 2\mathcal{H}(\Phi + \Psi)\}]}_{(+)}$$

$$= \partial_i \Psi' - \partial_i \Psi$$

$$(*) \rightarrow = \partial_i \Psi \mathcal{H}$$

$$(\dagger) \rightarrow = -\mathcal{H} \partial_i \Psi$$

Thus all contributions from $\lambda = 0$ cancel.

$$\underline{\underline{\text{Total} = 0}}$$

Terms from $\lambda = k$:

$$\partial_k \Gamma_{0i}^k - \partial_i \Gamma_{0k}^k + \underbrace{\Gamma_{kp}^k \Gamma_{0i}^p - \Gamma_{0k}^p \Gamma_{ip}^k}_{\substack{+ [\Gamma_{k0}^k \Gamma_{0i}^0 - \Gamma_{0k}^0 \Gamma_{i0}^k \\ + \Gamma_{kj}^k \Gamma_{0i}^j - \Gamma_{0k}^j \Gamma_{ij}^k]}}$$

$$= \partial_k ((\mathcal{H} - \Phi') \delta_i^k) - \partial_i ((\mathcal{H} - \Phi') \delta_k^k)$$

$$+ ((\mathcal{H} - \Phi') \delta_k^k) (\partial_i \Psi) - (\partial_k \Psi) ((\mathcal{H} - \Phi') \delta_i^k)$$

$$(*) \left\{ \begin{aligned} &+ (-2\delta_{(k}^k \partial_{j)} \Phi + \delta_{kj} \partial^k \Phi) ((\mathcal{H} - \Phi') \delta_i^j) \\ &- ((\mathcal{H} - \Phi') \delta_k^j) (-2\delta_{(i}^k \partial_{j)} \Phi + \delta_{ij} \partial^k \Phi) \end{aligned} \right.$$

$$(*) = \left[-(\delta_k^k \partial_j + \delta_j^k \partial_k) \Phi + \delta_{kj} \partial^k \Phi \right] \cdot ((\mathcal{H} - \Phi') \delta_i^j)$$

$$+ \left[(\delta_i^k \partial_j + \delta_j^k \partial_i) \Phi + \delta_{ij} \partial^k \Phi \right] \cdot ((\mathcal{H} - \Phi') \delta_k^j)$$

$$= \left[-4\partial_j \Phi + \partial_j \Phi \right] \cdot ((\mathcal{H} - \Phi') \delta_i^j)$$

$$+ \left[(\delta_i^j \partial_j + \delta_j^j \partial_i) \Phi + \delta_{ij} \partial^j \Phi \right] \cdot (\mathcal{H} - \Phi')$$

$$= 0$$

$$\downarrow$$

$$= 2\partial_i \Phi' + 2\mathcal{H}(\partial_i \Psi) = 2\partial_i (\Phi' + \mathcal{H}\Psi)$$

R_{ij} :

$$R_{ij} = \partial_\lambda \Gamma_{ij}^\lambda - \partial_j \Gamma_{i\lambda}^\lambda + \Gamma_{\lambda\rho}^\lambda \Gamma_{ij}^\rho - \Gamma_{i\lambda}^\rho \Gamma_{j\rho}^\lambda$$

Terms from $\lambda = 0$:

$$= \partial_0 \Gamma_{ij}^0 - \partial_j \Gamma_{i0}^0 + \underbrace{\Gamma_{0\rho}^0 \Gamma_{ij}^\rho - \Gamma_{i0}^\rho \Gamma_{j\rho}^0}_{\substack{+ \left[\Gamma_{00}^0 \Gamma_{ij}^0 - \Gamma_{i0}^0 \Gamma_{j0}^0 \right. \\ \left. + \Gamma_{0k}^0 \Gamma_{ij}^k - \Gamma_{i0}^k \Gamma_{jk}^0 \right]}}$$

$$= \partial_0 \left[\mathcal{H} \delta_{ij} - \{ \Phi' + 2\mathcal{H}(\Phi + \Psi) \} \delta_{ij} \right] \\ - \partial_j (\partial_i \Psi) + (\mathcal{H} + \Psi') (\mathcal{H} \delta_{ij} - \{ \Phi' + 2\mathcal{H}(\Phi + \Psi) \} \delta_{ij}) \\ - (\partial_i \Psi) (\partial_j \Psi) + (\partial_k \Psi) (-2\delta_{ik} \partial_j \Phi + \delta_{ij} \partial^k \Phi) \\ - ((\mathcal{H} - \Phi') \delta_i^k) (\mathcal{H} \delta_{jk} - [\Phi' + 2\mathcal{H}(\Phi + \Psi)] \delta_{jk}) \\ = [\mathcal{H}' - \Phi'' - 2\mathcal{H}'(\Phi + \Psi) - 2\mathcal{H}(\Phi' + \Psi')] \delta_{ij} - \partial_i \partial_j \Psi \\ + [\mathcal{H}^2 + \Psi' \mathcal{H} - \mathcal{H} \Phi' - 2\mathcal{H}^2(\Phi + \Psi)] \delta_{ij} \\ + [-\mathcal{H}^2 + 2\mathcal{H} \Phi' + 2\mathcal{H}^2(\Phi + \Psi)] \delta_{ij} \quad (\delta_i^k \delta_{jk} = \delta_{ij})$$

$$= [\mathcal{H}' - \Phi'' - 2\mathcal{H}'(\Phi + \Psi) - \mathcal{H}(\Phi' + \Psi')] \delta_{ij} - \partial_i \partial_j \Psi$$

Terms from $\lambda = k$:

$$\begin{aligned}
 &= \partial_k \Gamma_{ij}^k - \partial_j \Gamma_{ik}^k + \underbrace{\Gamma_{kp}^k \Gamma_{ij}^p - \Gamma_{ik}^p \Gamma_{jp}^k}_{\text{}} \\
 &\quad + \left[\Gamma_{k0}^k \Gamma_{ij}^0 - \Gamma_{ik}^0 \Gamma_{j0}^k \right. \\
 &\quad \left. + \Gamma_{km}^k \Gamma_{ij}^m - \Gamma_{ik}^m \Gamma_{jm}^k \right]
 \end{aligned}$$

$$= \partial_k \left[-2\delta_{(i}^k \partial_{j)} \Phi + \delta_{ij} \partial^k \Phi \right]$$

$$- \partial_j \left[-2\delta_{(j}^k \partial_{k)} \Phi + \delta_{jk} \partial^k \Phi \right]$$

$$+ ((\mathcal{H} - \Phi') \delta_k^k) (\mathcal{H} \delta_{ij} - [\Phi' + 2\mathcal{H}(\Phi + \Psi)] \delta_{ij})$$

$$- (\mathcal{H} \delta_{ik} - [\Phi' + 2\mathcal{H}(\Phi + \Psi)] \delta_{ik}) ((\mathcal{H} - \Phi') \delta_j^k)$$

$$\begin{aligned}
 O(2) \left\{ \right. &+ (-2\delta_{(k}^k \partial_{m)} \Phi + \delta_{km} \partial^k \Phi) (-2\delta_{(i}^m \partial_{j)} \Phi + \delta_{ij} \partial^m \Phi) \\
 &- \dots
 \end{aligned}$$

$$= \partial_k \left[(\delta_j^k \partial_i + \delta_i^k \partial_j) \Phi + \delta_{ij} \partial^k \Phi \right]$$

$$- \partial_j \left[(\delta_j^k \partial_k + \delta_k^k \partial_j) \Phi + \delta_{jk} \partial^k \Phi \right]$$

$$+ [3\mathcal{H}^2 + 3\mathcal{H}\Phi' - 3\mathcal{H}\Phi' + 6\mathcal{H}^2(\Phi + \Psi)] \delta_{ij}$$

$$- [\mathcal{H}^2 - 2\mathcal{H}\Phi' - 2\mathcal{H}^2(\Phi + \Psi)] \delta_{ij}$$

$$= \partial_i \partial_j \Phi + \nabla^2 \Phi \delta_{ij} \quad (\partial_k \partial^k \Phi \delta_{ij}) = \nabla^2 \Phi \delta_{ij}$$

$$+ 2 [\mathcal{H}^2 - 2\mathcal{H}\Phi' - 2\mathcal{H}^2(\Phi + \Psi)] \delta_{ij}$$

Combining $\lambda=0, \lambda=\kappa$ contributions,

$$\text{Total: } R_{ij} = [\mathcal{H}' - \Phi'' + 2\mathcal{H}^2(\Phi + \Psi)$$

$$- 2\mathcal{H}'(\Phi + \Psi) - 5\mathcal{H}\Phi' + \nabla^2 \Phi$$

$$- \mathcal{H}\Psi'] \delta_{ij} + \partial_i \partial_j (\Phi - \Psi)$$

$$= [\mathcal{H}' - \Phi'' + \nabla^2 \Phi - 5\mathcal{H}\Phi' - \mathcal{H}\Psi'$$

$$- 2(\mathcal{H}' + 2\mathcal{H}^2)(\Phi + \Psi)] \delta_{ij}$$

$$+ \partial_i \partial_j (\Phi - \Psi)$$

Phew!

At least that's
over ;)